

Model-Based Software Design, A.Y. 2024/25

Assignment #1

Components of the working group (max 2 people)

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Item definition (Example)

One pedal controller

Purpose of this document

The purpose of this document is to be the input for the “Hazard Analysis and Risk Assessment” (HARA) needed to comply with the ISO26262 standard. To ensure safety, all activities of the safety life cycle have to be planned to avoid systematic failures.

Therefore, this document describes the assumption on the **one pedal acceleration/braking system** item you should develop.

An additional purpose of this document is to define and describe the item, its functionality, dependencies on, and interaction with, the driver, the environmental conditions, external measure, the boundary of the item and interfaces to other items as well as assumptions concerning other elements at the vehicle level. This document will handle the requirements and recommendations for establishing the definition of the item, including its functionality, interfaces, environmental conditions, legal requirements, and known hazards.

Purpose of the item

Please describe in this chapter the purpose of the item. Consider laws, standards, and regulations to sufficiently describe the item's purpose.

The purpose of the item is the following:

- To allow the driver to set the torque (positive→acceleration, negative→braking) applied on the driving wheels of a car. This system enables the driver to use only the throttle pedal for both the functions of accelerating or braking (up to a certain level) the vehicle. This system only allows use of the regenerative braking function of an electric/hybrid vehicle.
- As an assumption, the braking pedal is still inside the car, it acts directly on the hydraulic braking system, and its circuitry is independent of interferences from the “one” throttle/braking pedal. The information on whether the brake pedal is pressed is available for the considered item.

Functional behavior

The automatic transmission selector is implemented as a by-wire (hence, no mechanical links between the transmission and the selector are present) and features, in the order, these positions: P (park), R (reverse), N (neutral), D (drive), and B (braking/one pedal). The driver can move the transmission selector at any moment, so the actually selected mode is shown on the dashboard screen. The item switches to the position chosen by the driver as soon as all related safety conditions are met.

The system can adopt two different behaviours, one when the automatic transmission selector (an independent system) is in the D position and the other in the B.

In particular:

- In D position mode, it reads the position of the throttle pedal and requires a traction torque proportional to the pedal position, as traditional in the automotive market. When

the pedal is completely released, no torque is required meaning that the vehicle has its own braking force due to interaction with the air or the terrain, the internal combustion engine, or just the transmission power consumption due to internal frictions in the case of an electric vehicle. In this mode, to increase the braking torque, it is necessary to press the brake pedal and stop the vehicle completely. When the brake pedal is released in cars equipped with automatic transmissions, the vehicle starts to move slowly.

- In B (brake) position mode, the throttle pedal travel is divided into two regions:
 - regenerative braking, from the complete release up to a certain point (for example, 1/3 of the travel angle) that we can call the *neutral point*. The readout from the pedal inside this region is interpreted as a request for a braking torque, maximum when the pedal is completely released, then proportionally decreased upon the *neutral point*. When the pedal is released, the vehicle brakes up to completely stop its motion. From then on, the car remains stopped automatically regardless of the street slope. To make the vehicle moving, it is necessary to press the throttle pedal up to the acceleration region, described in the following, or to press the brake pedal and then release it.
 - Acceleration, from the neutral point up to the end of the travel (acceleration region), where the position is interpreted as a request of a traction torque proportional to the pedal position.

The behaviour can be described mathematically as follows:

$$\left\{ \begin{array}{ll} \tau_r = -\max(\tau_a) \cdot (1 - 3p), & \text{when } 0 < p \leq \frac{1}{3} \text{ (braking region).} \quad (1) \\ \tau_r = \max(\tau_a) \cdot \frac{3}{2} \cdot \left(p - \frac{1}{3}\right), & \text{when } \frac{1}{3} < p \leq 1 \text{ (acceleration region).} \quad (2) \end{array} \right.$$

where:

- τ_r is the requested torque;
- p is the pedal position expressed in normalized [0,1] range;
- $\max(\tau_a)$ is the maximum acceleration torque in the forward direction.

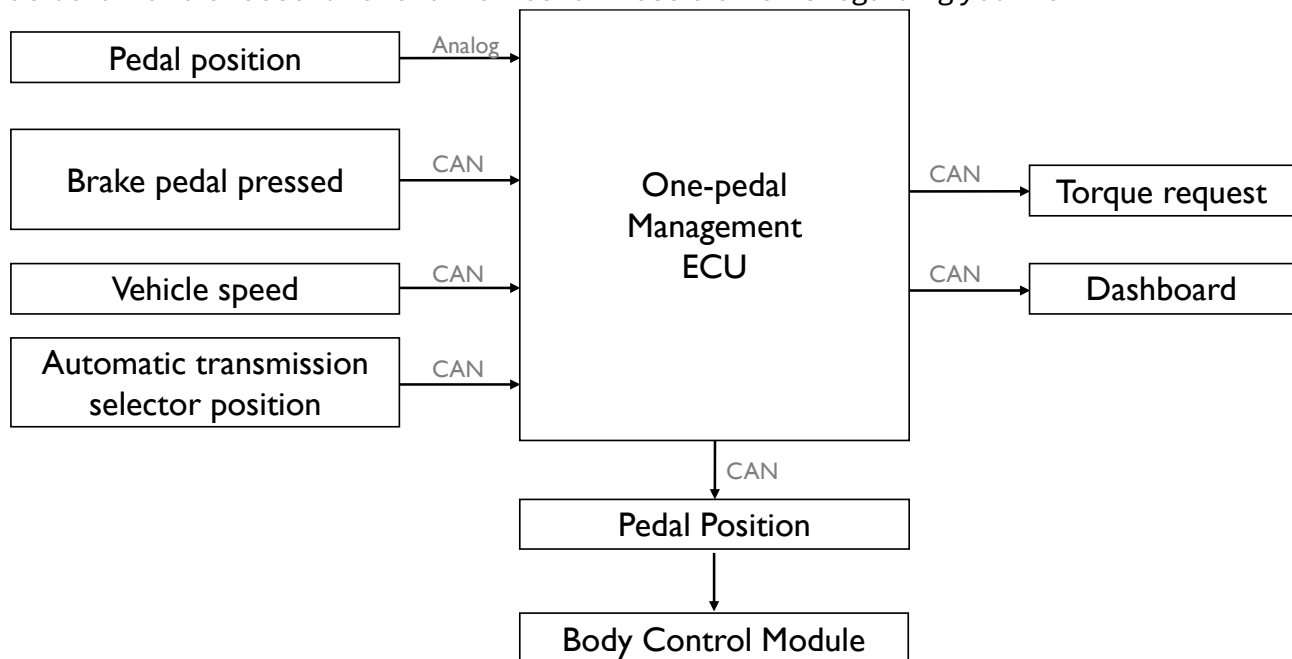
The requested torque is positive to indicate a forward acceleration action or negative to indicate a braking (or backward) acceleration action (from here, the – sign in the equation 1).

Of course, it is still possible to use the braking pedal in case of emergencies or to increase the braking torque thanks to the hydraulic brakes.

Function	Operating elements
Determine torque request	Throttle pedal
Select transmission mode (behavior)	Automatic transmission selector
Brake pedal pressed	Data from the CAN bus regarding the status of the braking pedal
Driver notifications	Tell the driver the selected mode on the dashboard (between P, R, N, D, and B) and eventual faults. This is usually the one chosen by the by-wire selector

Functional block diagram

Please describe the interaction with external systems or items and/or interfaces to other elements outside the boundary of your item. Please consider the combination of “sensor-logic-actuator” and choose functional names for these elements regarding your item.



Boundaries of the system responsibility and interfaces

Please describe the boundary of the system responsibility, interaction with external systems or items and interfaces to other elements outside your item in combination with the block diagram above

The system is in charge of providing the torque request (positive or negative) to the electric motor (EM) electronic control unit (ECU).

It provides this request through the vehicular Controller Area Network (CAN).

It has to compute this torque request based on the gear selector position (negative torque only for the B position).

Moreover, it has to check the vehicle speed to determine the torque effects, in particular preventing that, during the regenerative braking, the negative torque request causes the vehicle to move in the reverse direction.

Another responsibility is to keep the vehicle stopped until the throttle pedal reaches the acceleration position and to monitor when the braking pedal is pressed to make the car slowly move when it is released.

In the reverse gear, the car acts like a standard automatic transmission car, so the vehicle only stops when the braking pedal is pressed and starts to slowly move backward when it is released.

Moreover, the transition between the position N and R or D/B is accepted only when the speed of the vehicle is lower than 5 km/h (in the same motion direction) AND the brake pedal is pressed, with the only exception on the selection of the N (neutral), which can be accepted at any time and causes the vehicle to move freewheel. The transition between R and P can be accepted only with the car almost still, and the braking pedal pressed.

Other sources of hazards, which influence the safety and reliability of the item

Please describe other sources (not E/E) of hazards, which influence the safety and reliability of the item

Functional requirements

Please describe all already noted functional safety requirements, this is normally output of H&R.

Other requirements

Other environmental requirements which can influence your item

Law, directive and standard

List the laws, directives and standard which have to be considered

External measure to minimizing risks

Which external measures can be taken in order to minimize the risk:

Hazard Analysis and Risk Assessment (Example)

One pedal controller

Participants

Name, department	Qualification	Experience
Federica D'Ugo, 345772		
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Analyzes of Hazardous Situations

H1	Unintended vehicle acceleration
H2	Unintended vehicle breaking
H3	Insufficient vehicle acceleration
H4	Insufficient vehicle breaking
H5	Unintended vehicle motion in incorrect direction

H1

The vehicle accelerates without driver input, possibly due to sensor failures, software errors, or CAN interference. This may lead to collisions in traffic or confined spaces.

H2

The vehicle brakes unexpectedly due to misinterpretation of throttle pedal input or regenerative braking malfunction. Sudden deceleration can cause rear-end collisions.

H3

The vehicle does not accelerate as expected, potentially due to sensor failures, software errors, or power limitations. This may cause rollback on hills or failed overtaking maneuvers.

H4

The vehicle does not decelerate as expected when braking is required. This may be caused by a failure in the regenerative braking system, potentially resulting in longer stopping distances.

H5

The vehicle moves in an unintended direction due to software errors, sensor data transmission issues, or incorrect gear selection. This can lead to accidents during parking or starting on slopes.

Analyses of situations

Definition of possible functional failures

Failure #	Description
F1: Throttle pedal sensor failure	<i>The system cannot correctly interpret the driver's input, leading to unintended acceleration</i>

F2: Brake pedal communication failure	<i>The system fails to receive brake pedal input, preventing transitions between gear positions (N to R, N to D, N to B, or R to P) and possibly failing to keep the vehicle stationary</i>
F3: Communication failure with the automatic transmission selector	<i>The system does not receive the updated transmission mode selection, preventing a correct transition to the newly selected gear.</i>
F4: Communication failure with vehicle speed sensor	<i>The system does not receive vehicle speed data, which prevent transitions between gear positions such as N/R, N/D, or N/B.</i>
F5: Communication failure with the dashboard interface	<i>A failure to send data to the dashboard may prevent the driver from seeing the correct transmission mode or system status.</i>
F6: Incorrect torque request computation	<i>The system calculates an incorrect torque request, resulting in acceleration that does not match the driver's intention or the actual sensor inputs.</i>

Driving scenarios

Describe the possible driving situations and define the status of the vehicle you want to consider

Description of the possible driving situations

Definition of the vehicle status

Considerations

Describe driving situations for each status of the vehicle.

Operational Situations #	Driving situation	Vehicle status
S1	City driving (50 km/h)	The vehicle is accelerating or maintaining speed
S2	Temporarily stopped (for example at a red light)	The vehicle is stopped with engine on, ready to move
S3	Parked	The vehicle is stopped with engine off
S4	Reverse driving	The vehicle is moving in the opposite direction of normal travel, at low speed (30 km/h)
S5	Braking with the brake pedal	The vehicle is at high speed and needs to slow down fast
S6	Deceleration with regenerative braking	The vehicle is slowing down without using the brake pedal
S7	High way driving	The vehicle moves at high speed

Scenario 3 and Scenario 4 are not considered in the estimation matrix, as we have decided to focus exclusively on scenarios involving the B and D driving modalities.

Analysis

Estimation matrix

		Risk parameters	Scenario 1	Scenario 2	Scenario 5	Scenario 6	Scenario 7	TOP EVENT
HAZARDS	H1	S	3	1	3	2	1	C
		E	4	4	1	4	4	
		C	2	1	0	1	1	
		ASIL	9	6	4	7	6	
	H2	S	2	0	0	1	3	C
		E	4	4	1	4	4	
		C	1	0	0	1	2	
		ASIL	7	4	1	6	9	
	H3	S	1	0			2	A
		E	4				4	
		C	1				1	
		ASIL	6	0	0	0	7	
	H4	S	2			2	3	B
		E	4		0	4	4	
		C	2			1	1	
		ASIL	8	0	0	7	8	
	H5	S		1		2		A
		E		4	1	4		
		C		1		1		
		ASIL	0	6	1	7	0	

Scenario 5 has an exposure value of 1, as it represents an exceptionally rare occurrence.

In contrast, the other scenarios have an exposure value of 4 due to their high frequency.

- **Scenario 1** (city driving) with **H5** (unintended vehicle motion in incorrect direction): when using driving modes *D* or *B*, it is not possible for the vehicle to suddenly move in an unintended direction. Since the two facts are inconsistent, the hazard is considered **Not Applicable**.
- **Scenario 2** (temporarily stopped) with **H4** (insufficient vehicle braking): as the vehicle is already stationary, no further braking is required. Therefore, the hazard is **Not Applicable**.
- **Scenario 6** (deceleration with regenerative braking) with **H3** (insufficient vehicle acceleration): during regenerative braking, no acceleration is expected. Therefore, this hazard is **not consistent** with the scenario.
- **Scenario 7** (highway driving) with **H5** (unintended vehicle motion in incorrect direction): at high speeds, it is practically impossible for the vehicle to suddenly change direction unintentionally. The two elements are incompatible, and the hazard is **Not Applicable**.
- **Scenario 5** (braking with the brake pedal) with **H3** (insufficient vehicle acceleration): is also **not applicable** because when the objective is to stop, acceleration is unnecessary.

- **Scenario 2** (temporarily stopped) with **H3** (insufficient vehicle acceleration): in this situation, the vehicle is expected to accelerate when the light turns green or when the driver intends to resume driving. However, even in the case of insufficient acceleration, this condition alone is unlikely to cause any injury. Therefore, the **Severity is rated 0**, and the scenario is considered not safety relevant.
- **Scenario 5** (braking with the brake pedal) with **H4** (insufficient vehicle braking): in this case, the **Exposure is rated 0** because the brake pedal is considered an external mechanical component, assumed to always be functional. Therefore, it is highly unlikely that insufficient braking would occur, especially when using the emergency brake.

Operational Situations – Comment of entries

Start with the description of what happens and then assign the parameters.

Please analyze in this way two other scenario/failure associations at your choice.

Effect	<i>Driving uphill and insufficient acceleration</i>	
Statement S	<i>If the vehicle doesn't provide sufficient acceleration while driving uphill, there is a significant risk of rolling backward. This could cause a collision with the vehicle behind, leading to light to moderate injuries depending on the speed and the nature of the impact. The risk of harm is high in stop and go traffic, especially on slopes, where nearby vehicles leave little room for error.</i>	S 1
Statement E	<i>This situation occurs occasionally, typically during stop-and-go traffic on uphill gradients. The frequency of this event is estimated to occur between 1% and 10% of the total driving time, with a higher probability in urban areas or steep hill climbs.</i>	E 3
Statement C	<i>The driver can press the brake pedal to prevent the vehicle from rolling backward. However, this may not be enough if the driver does not react quickly.</i>	C 1

Effect	<i>Driving downhill and unintended acceleration</i>	
Statement S	<i>If the vehicle unexpectedly accelerates while driving downhill, it may gain speed rapidly and become difficult to control. The slope increases the effect of the acceleration, raising the risk of colliding with another vehicle or a pedestrian, potentially resulting in serious or fatal injuries, especially if the driver doesn't react quickly.</i>	S 3
Statement E	<i>This type of event may occur occasionally, particularly if the system malfunctions due to sensor or software issues. The frequency of this scenario is estimated at 1% to 10% of the time, with a higher risk during</i>	E 3

	<i>downhill driving in areas where vehicle speed may not be adequately monitored.</i>	
Statement C	<i>The driver can press the brake pedal to slow down the car and potentially stop the unintended acceleration. In vehicles with regenerative braking, this may also help to bring the car to a halt. However, if the system malfunction is severe, braking alone might not be enough to control the vehicle, and additional measures may be needed.</i>	C 1

Safety goals

SG1	The item should avoid unintended breaking
SG2	The item should avoid unintended acceleration
SG3	The item should avoid insufficient acceleration
SG4	The item should avoid insufficient breaking
SG5	The item should avoid motion in incorrect direction

Results

Failure/ Malfunction/ Hazardous situation	Safety goal	ASIL-level (highest)	Safe state	Fault tolerance time interval (FTTI)
Unintended vehicle breaking	SG1	C	The item should switch to D mode and properly inform the driver through the dashboard	0.5/1 seconds
Unintended vehicle acceleration	SG2	C	The system responds to unintended acceleration by setting the torque to zero, switching to Neutral mode, and clearly informing the driver via the dashboard.	0.5/1 seconds
Insufficient vehicle acceleration	SG3	A	The system should inform the driver of the insufficient acceleration issue through the dashboard	1/2 seconds
Insufficient vehicle breaking	SG4	B	The system sets the torque to zero, switches to Neutral mode, and properly informs the driver via the dashboard	1/2 seconds
Unintended vehicle motion in incorrect direction	SG5	A	The system requests zero torque and clearly informs the driver through the dashboard	0.5 seconds

Relevant failure modes for H1: Unintended vehicle acceleration

Relevant failure modes for H1	Description	Reason for Relevance
F1	Throttle pedal sensor failure	If the system misunderstands the driver's input, the car might speed up when the driver doesn't want it to.
F6	Incorrect torque request computation	An incorrect torque calculation can cause the vehicle to accelerate when the driver doesn't intend to.

Relevant failure modes for H2: Unintended vehicle braking

Relevant failure modes for H2	Description	Reason for Relevance
F1	Throttle pedal sensor failure	If the system falsely detects a throttle release, it might suddenly reduce torque, simulating braking.
F6	Incorrect torque request computation	An incorrect torque request can lead to unintended deceleration.